

2 A student grew a bacterial culture in a liquid medium for 20 hours.

Every hour, a sample of the culture was removed.

This sample was added to some fresh medium and mixed thoroughly.

This mixture was spread onto agar in a Petri dish, so that the number of bacteria in the culture could be determined.

(a) Which method is being used to determine the number of bacteria in this culture?

(1)

- ☐ **A** dilution plating
- ☐ **B** latency
- ☐ **C** mass method
- ☐ **D** optical method (turbidity)

(b) Explain how the Petri dish should have been sealed after the mixture was spread onto the agar.

(2)

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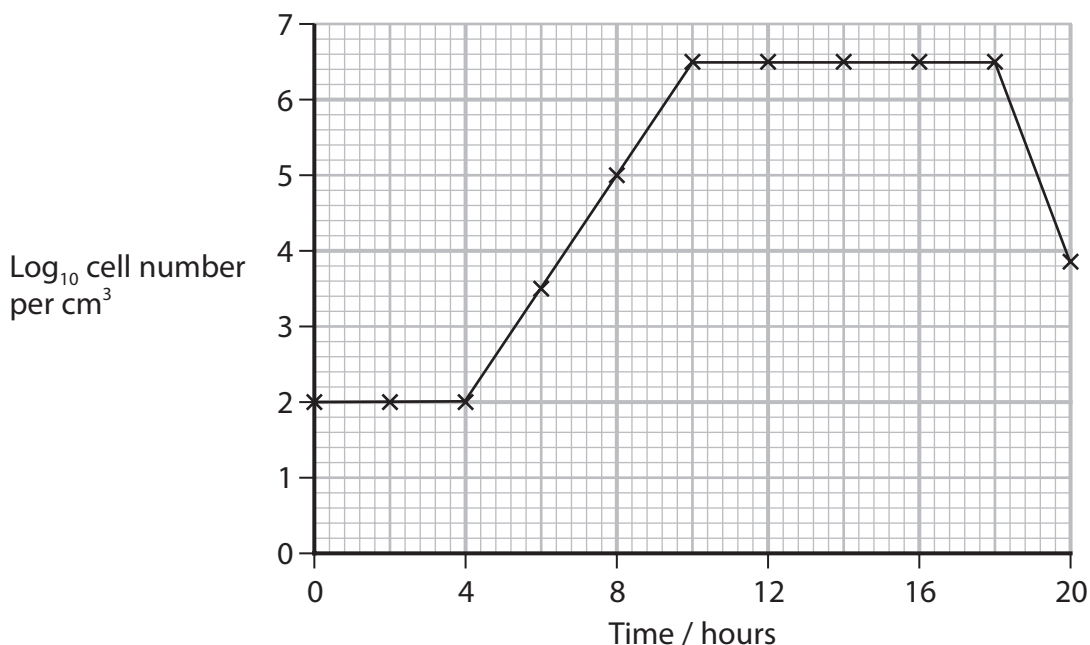
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(c) The graph shows the bacterial growth curve for this culture.



(i) Which shows the order of the phases of a bacterial growth curve?

(1)

- ☐ **A** exponential → lag → death → stationary
- ☐ **B** lag → stationary → death → exponential
- ☐ **C** lag → exponential → stationary → death
- ☐ **D** stationary → exponential → lag → death

(ii) The bacteria were not growing at their fastest rate in the conditions the student used.

How many of the following changes could result in a faster growth rate?

- decrease in pH
- decrease in temperature
- increase in pH
- increase in temperature

(1)

- ☐ **A** 1
- ☐ **B** 2
- ☐ **C** 3
- ☐ **D** 4

(iii) Calculate the growth rate constant (k) of these bacteria between 4 hours and 10 hours.

(2)

Use the equation: $k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$

Give your answer to two decimal places.

Answer

(Total for Question 2 = 7 marks)